Ahmed Reda Alnajjar

Full-Stack Web Developer | Programming & AI Instructor

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PROFESSIONAL SUMMARY

Computer Science graduate (Canadian International College) who builds full-stack web products and teaches programming and AI. Built and shipped Lokeify, a multi-store e-commerce platform with a store builder, merchant dashboard and 3D fit room, plus 15+ other projects spanning React, Next.js, Node.js, Firebase, React Native and AI speech-assessment tools. At Zoomix I write the curricula other instructors teach from (a 24-session full-stack web course, Scratch and AI tracks), teach kids, teens and adults, and build the internal systems staff use daily. ICPC participant; DEPI React Frontend graduate; CS50 AI (Harvard).

TECHNICAL SKILLS

Languages	JavaScript (ES6+), TypeScript, Python, Java, SQL, HTML5, CSS3, Sass
Frontend	React.js, Next.js (App Router), Vite, Tailwind CSS, Bootstrap, Framer Motion, Zustand, Three.js / React Three Fiber, responsive UI
Mobile	React Native, Expo Router
Backend & Data	Node.js, Express, Next.js Route Handlers, FastAPI, Flask, REST APIs, session auth, Firebase (Auth, Firestore, Storage), MongoDB / Mongoose, SQLite / Turso
AI & ML	Scikit-learn, Pandas, NumPy, TF-IDF, classification & model evaluation, LLM APIs (OpenAI, Claude, Gemini, Groq), Azure Speech pronunciation assessment, prompt engineering
AI Automation	n8n, Make, Zapier, ChatGPT, Claude, Gemini, AI agents & chatbots
Tools & Cloud	Git, GitHub, Vercel, Render, Cloudinary, Postman, Google Colab, Jupyter, VS Code, NetBeans
Teaching	Curriculum & lab design, live online sessions, project-based lessons, assessments, instructor mentoring; Scratch, ScratchJr, App Lab, Minecraft coding, mBlock, PictoBlox

EXPERIENCE

Technical Curriculum & Internal Systems Lead — Zoomix — Cairo, Egypt · 2026 – Present

- Wrote a 24-session full-stack Web Development curriculum (HTML, CSS, JavaScript, DOM & APIs, Node.js, Express, REST, MongoDB/Mongoose, MVC, authentication, testing & deployment) with slides, an instructor guide and a runnable project for every session.
- Prepared course material for the Scratch (24 lectures), App Lab, AI Fundamentals and AI & Automation tracks for kids, teens and adults.
- Teach live programming and AI sessions to kids, teens and adult beginners, and AI-automation workshops (ChatGPT, Claude, Gemini, n8n, Make, Zapier, chatbots and agents) for university professors and business owners.
- Built internal web systems used daily by staff — a real-time lecturer timetable & instructor management system (zoomix-lec.vercel.app) and a sales pipeline tool — covering UI, API and database design.
- Train and mentor the instructor team through session rehearsals, code reviews and debugging support; built AI content pipelines (LLM APIs) that generate course material and video scripts.

Coding Instructor — iSchool — Online · 2025 (6 months)

- Taught live online coding classes to kids and teens (ages 6–18) across the Arab region: Scratch, Minecraft coding, web development (HTML, CSS, JavaScript) and AI.
- Taught AI fundamentals, algorithms and the Software Development Life Cycle to grades 4–6, and mentored students building games and projects in mBlock and PictoBlox.

PROJECTS

Lokeify — Multi-Store E-commerce Platform (Store Builder + Merchant Dashboard) — 2026

Next.js, React, TypeScript, Tailwind CSS, Three.js / React Three Fiber, Zustand, SQLite / Turso, Vercel · Live: lokeify.vercel.app · GitHub: github.com/ahmed-reda-elnajjar/lokeify

- Store builder: merchants sign up, open a store and get their own live storefront; one codebase serves every store with each query scoped to its shop.
- Theme & sections editor with a draft → publish flow; product editor (photos, colorways, sizes, stock), orders, customers and a 30-day sales dashboard.
- Storefront with collections, search, bag, checkout (server-side pricing and stock deduction), customer accounts, a 3D fit room, size advice and AI try-on.
- Production basics: signed http-only cookie sessions, scrypt passwords, encrypted per-store API keys and chunked uploads on Vercel.

CRATE Store — Streetwear Brand Storefront 2026

Next.js, React, TypeScript, Three.js, Vercel · Live: crate-store.vercel.app · GitHub: github.com/ahmed-reda-elnajjar/crate-store

- Storefront for the CRATE streetwear brand, built from a design handoff: timed drops with countdowns, shop, product pages, lookbook, search, bag, checkout, customer accounts and an admin area; also serves as the live demo store inside Lokeify.

MarketScan Pro — Smart Supermarket System (Graduation Project) 2025

React.js, Tailwind CSS, Web Speech API, SVG · React Native / Expo (mobile) · Node.js (server edition) · Live: marketscan-pro.vercel.app · GitHub: github.com/ahmed-reda-elnajjar/MarketScan-Web · Mobile app: github.com/ahmed-reda-elnajjar/marketscan-shopper-app

- Three role-based interfaces (Shopper, Employee, Admin): SVG store map with multi-stop shortest-path routing, voice search (Web Speech API) and a Smart Cart that tracks budget and suggests cheaper alternatives.
- Employee portal with task procedures and simulated barcode/QR scanning; Admin panel with inventory CRUD and a heatmap of high-traffic zones; companion React Native (Expo) mobile app.

Peaky Scouts & GTX — Call-Center Recruitment Platforms 2024 – 2026

React.js, Tailwind CSS, Firebase (Auth, Firestore, Storage), Cloudinary, Framer Motion, Three.js, Gemini API, Vercel · Live: peaky-scouts.vercel.app · GitHub: github.com/ahmed-reda-elnajjar/Egypt_Hire · GTX: gtx-rho.vercel.app

- Hiring platforms connecting call-center agents and employers: real-time job listings, application flow with voice-recording screening and CV upload, recruiter form and an admin dashboard for jobs and candidates.
- GTX adds English-course pages for applicants (Career Sprint), recommended jobs and user profiles; both shipped through continuous deployment on Vercel.

Callify — AI English Speaking Assessment for Call-Center Applicants 2026

React.js, Azure Speech SDK (pronunciation assessment), Groq LLM, Firebase, Cloudinary · GitHub: github.com/ahmed-reda-elnajjar/callify

- Scores recorded answers with phoneme-level pronunciation, fluency and completeness from Azure Speech, analyses grammar and vocabulary with an LLM, and maps the result to a CEFR level (A1–C1) with strict thresholds.
- Moved all API keys to environment variables, added score-bounds checks, weighted segment averaging and 30-second timeouts.

English Assessment Engines — Three AI Speaking-Assessment Back Ends 2026

FastAPI · Express · Flask · Azure Speech, Google STT, Deepgram · Claude, GPT, Groq · GitHub: github.com/ahmed-reda-elnajjar/english-assessment · Express: github.com/ahmed-reda-elnajjar/english-evaluator · Flask: github.com/ahmed-reda-elnajjar/ai-english-assessment

- Built the same CEFR speaking assessment three ways to compare stacks and providers: browser recording/upload → audio conversion (ffmpeg) → speech scoring → LLM grammar/vocabulary feedback → score ring and radar-chart report.

Zoomix Lecturer Timetable — Instructor Scheduling System 2026

React, Vite, Tailwind CSS, Firebase Firestore (real-time), Framer Motion · Live: zoomix-lec.vercel.app · GitHub: github.com/ahmed-reda-elnajjar/zoomix-LEC

- Weekly lecture timetable with live status per instructor, real-time Firestore sync and add/remove sessions; Arabic RTL interface used by Zoomix staff.

Save Time — English Course Platform 2026

HTML, CSS, JavaScript · Node.js, Express, SQLite · Live: savetime-english.vercel.app · GitHub: github.com/ahmed-reda-elnajjar/savetime_english_course-

- Course site with course details, testimonials, auth, student dashboard and admin, backed by an Express REST API (auth, courses, enrollments, contacts, videos) on SQLite.

AI-Based Text Classifier — End-to-End ML Pipeline 2024

Python, Scikit-learn, Pandas, NumPy, Google Colab

- Cleaning, TF-IDF features, training and evaluation; benchmarked five classifiers on imbalanced text data using precision, recall and F1.

Bubble Pop — Hand-Controlled Webcam Game 2026

JavaScript, Canvas API, getUserMedia · Live: bubble-pop-game-zeta.vercel.app · GitHub: github.com/ahmed-reda-elnajjar/bubble-pop-game

- Tracks the player's hand from live webcam frames (with smoothing and adjustable sensitivity) to steer a cursor and pop bubbles; built as a teaching demo for kids.

Connect Four AI Agents — Python 2025

Python, OOP, Jupyter / Colab · GitHub: github.com/ahmed-reda-elnajjar/connect-four-ai

- Connect Four engine with win/draw detection and a pluggable Agent class so different AI strategies can play each other.

DEPI Front-End Projects — Negroo Move, Trafalgar, Lookscout and more 2024

HTML5, *CSS3*, *Sass*, *Bootstrap*, *JavaScript* · Live: depi-web-projects.vercel.app · GitHub: github.com/ahmed-reda-elnajjar/depi-web-projects · Negroo Move: negroo-move.vercel.app

- Responsive sites built in the DEPI React Frontend track: Negroo Move (movie platform with sign-up, favorites and team pages), Trafalgar (healthcare landing page), Lookscout, Connect Four, To-Do list and more.

Java Swing Desktop Apps — Student Registration Wizard & Drawing App 2025

Java, *Swing*, *OOP* · GitHub: github.com/ahmed-reda-elnajjar/java-swing-projects

- Multi-screen registration wizard (interface-based controller, abstract panels, CardLayout) and a drawing tool with shape classes, colour selection and freehand strokes.

EDUCATION

B.Sc. in Computer Science — Canadian International College (CIC), New Cairo Graduated 2026

- ICPC participant (International Collegiate Programming Contest) — competitive programming and algorithmic problem-solving.
- Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming (Java), Databases, Software Engineering, Mobile Programming, Information Security, Embedded Systems, Computer Networks, Linear Algebra, Probability & Statistics.

CERTIFICATIONS & COURSES

React Frontend Web Developer 2024
Digital Egypt Pioneers Initiative (DEPI), Ministry of Communications & IT — 8-month professional track

Business English Track Apr – Oct 2024
DEPI & Berlitz Egypt, Ministry of Communications & IT

CS50's Introduction to Artificial Intelligence with Python 2024
Harvard University (online) — search, knowledge, optimization, neural networks, language models

Project Management Course Aug 2024
British University in Egypt (BUE) — Farouk El-Baz Sustainability Research Centre

Introduction to Artificial Intelligence & Applications Aug 2023
Zewail City of Science and Technology — 50-hour blended program

Entrepreneurship Jan 2025
Canadian International College (CIC) with IGS & Partners

LANGUAGES & SOFT SKILLS

Languages: Arabic — Native · English — Fluent (professional working proficiency)

Soft skills: Explaining technical ideas clearly, mentoring, teamwork, structured problem-solving, time management, adaptability